

[CSE445(5), Summer 2026] Assignment-1: Multivariate Regression

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I. INTRODUCTION

Superconductivity is a phenomenon in which materials exhibit zero electrical resistance below a specific critical temperature. Predicting this temperature is important for developing advanced technologies such as energy-efficient power systems, magnetic levitation, and quantum computing. Since experimental discovery of superconducting materials is both costly and time-consuming, machine learning offers an efficient alternative by learning patterns from existing data to estimate the critical temperature of new materials.

In this study, a Linear Regression model is developed using selected features from a superconductivity dataset. To improve the model's ability to capture nonlinear relationships, polynomial feature engineering is applied by introducing squared features. The performance of the baseline and enhanced models is evaluated using Mean Squared Error (MSE) and the coefficient of determination (R^2).

The main contributions of this work are summarized as follows:

- **Development of a Linear Regression model for predicting the critical temperature of superconducting materials.**
- **Application of feature selection and robust scaling to improve data preprocessing and model training.**
- **Investigation of polynomial feature engineering through the addition of squared features.**
- **Comparative evaluation of the baseline and polynomial models using MSE, R^2 , convergence analysis, and prediction visualization.**

II. METHOD

A. Data Preparation and Pre-processing

1) *Splitting the Dataset:* We randomize the data and use 75% of the data to train the model and the remaining 25% for testing.

```
1 train_set = df.sample(frac=0.75, random_state=445)
2 test_set  = df.drop(train_set.index)
```

2) *Feature Engineering:* Since this is a standard regression problem with a very aggregated dataset, extensive feature extraction may not be necessary. However, the dataset contains redundant variables for example, `mean_atomic_mass` and `wtd_mean_atomic_mass` carry overlapping information, making it unnecessary to include both simultaneously, so we discard some features as necessary and only keep 20 of the most representative ones.

3) *Data Standardization:* We apply **Robust Scaling** to the training dataset to ensure smoother gradients and stable convergence during model training. Robust scaling is preferred here because it uses the median and interquartile range (IQR) rather than the mean and standard deviation, making it less sensitive to outliers.

```
1 # Scaling the dataset
2 median = x_train.median()
3 q1      = x_train.quantile(0.25)
4 q3      = x_train.quantile(0.75)
5 iqr     = q3 - q1
6
7 x_train_scaled = (x_train - median) / iqr
8 x_test_scaled  = (x_test  - median) / iqr
```

B. Method Selection

1) *Supervised Learning:* We perform a multivariate linear regression using Gradient Descent as our main iterative optimization algorithm. Since it involves training algorithms on labeled datasets, i.e., input-output pairs.

C. Training and Optimization

The chosen algorithm is fed the training data. During this iterative process, the model analyzes the features and learns the underlying patterns by minimizing its prediction errors.

III. EXPERIMENT RESULTS

This section presents the performance evaluation of the proposed Linear Regression model for predicting the critical temperature of superconducting materials. Two different experiments were conducted to investigate the impact of feature engineering on model performance.

The first experiment utilized the selected 20 features obtained from the dataset, while the second experiment incorporated polynomial feature expansion by adding squared terms for each selected feature. Both models were trained using Gradient Descent optimization and evaluated using the Mean Squared Error (MSE) and the coefficient of determination (R^2).

A. Experimental Setup

The superconductivity dataset was first preprocessed by handling missing values and selecting twenty relevant features based on their statistical importance. The dataset was divided into training and testing sets using a 75:25 split.

To ensure numerical stability during optimization, Robust Scaling was applied to the selected features before training the model.

For the baseline Linear Regression model, the following hyperparameters were used:

- **Learning Rate: 0.001**
- **Number of Iterations: 2000**
- **Optimizer: Gradient Descent**

For the polynomial feature model, squared terms of all selected features were added, and the model was trained using:

- **Learning Rate: 0.01**
- **Number of Iterations: 5000**
- **Optimizer: Gradient Descent**

B. Training Convergence

The cost function was monitored throughout the optimization process to evaluate the convergence behavior of Gradient Descent.

Figure 1 illustrates the cost reduction obtained using the original selected features.

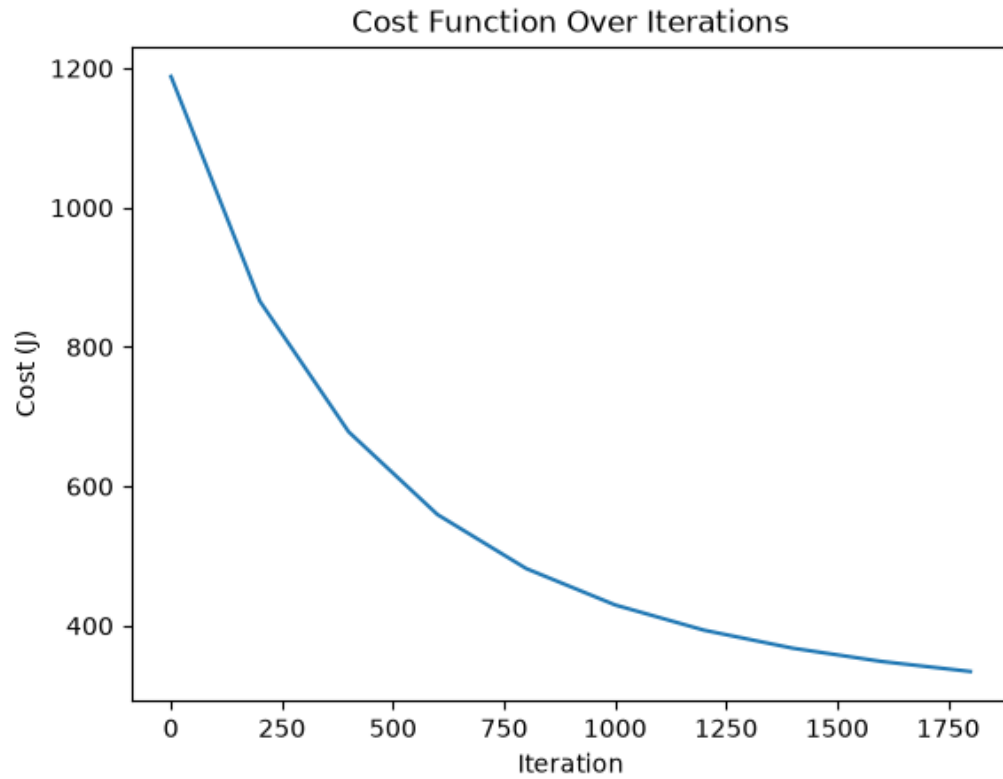


Fig. 1: Cost function versus iterations using the selected features.

The initial cost of the model was approximately 1188.25. As the optimization progressed, the cost continuously decreased and converged to approximately 334 after 2000 iterations, indicating successful minimization of the objective function.

After introducing squared features, the convergence behavior improved significantly, as shown in Figure 2.

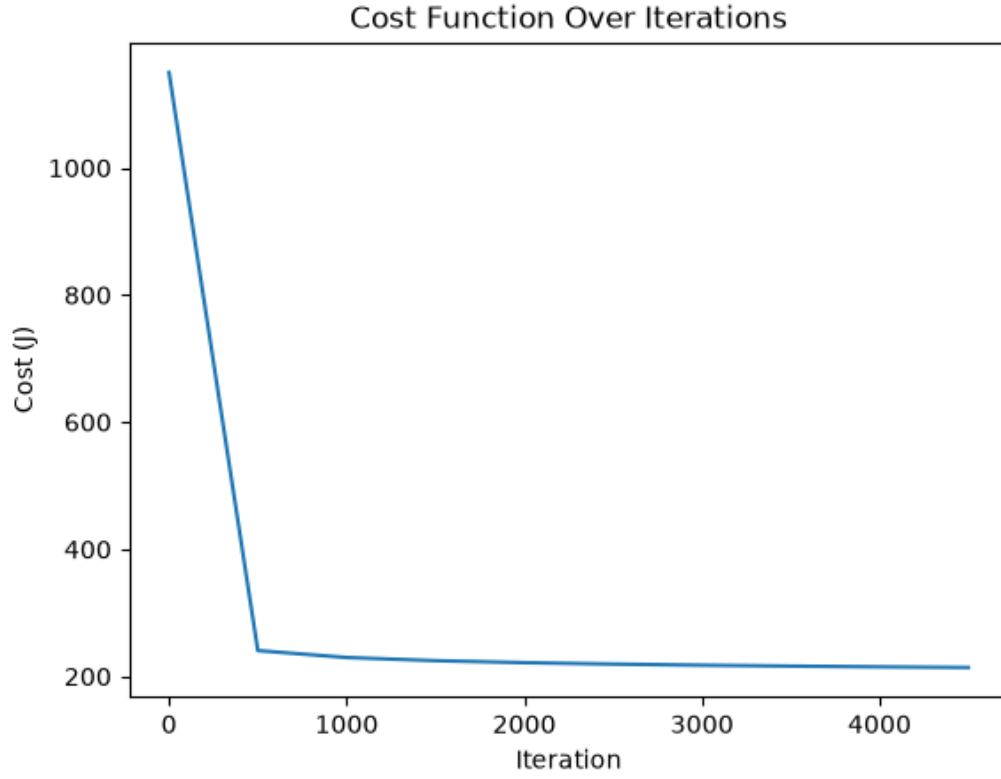


Fig. 2: Cost function versus iterations after adding squared features.

The polynomial model started with an initial cost of approximately 1149.99 and rapidly converged to nearly 214 after 5000 iterations. The lower final cost demonstrates that the additional nonlinear features enable the model to fit the underlying relationship between the input variables and critical temperature more effectively.

C. Prediction Performance

The prediction capability of both models was evaluated by comparing the predicted critical temperatures against the actual values from the testing dataset.

Figure 3 shows the prediction results obtained using the original feature set.

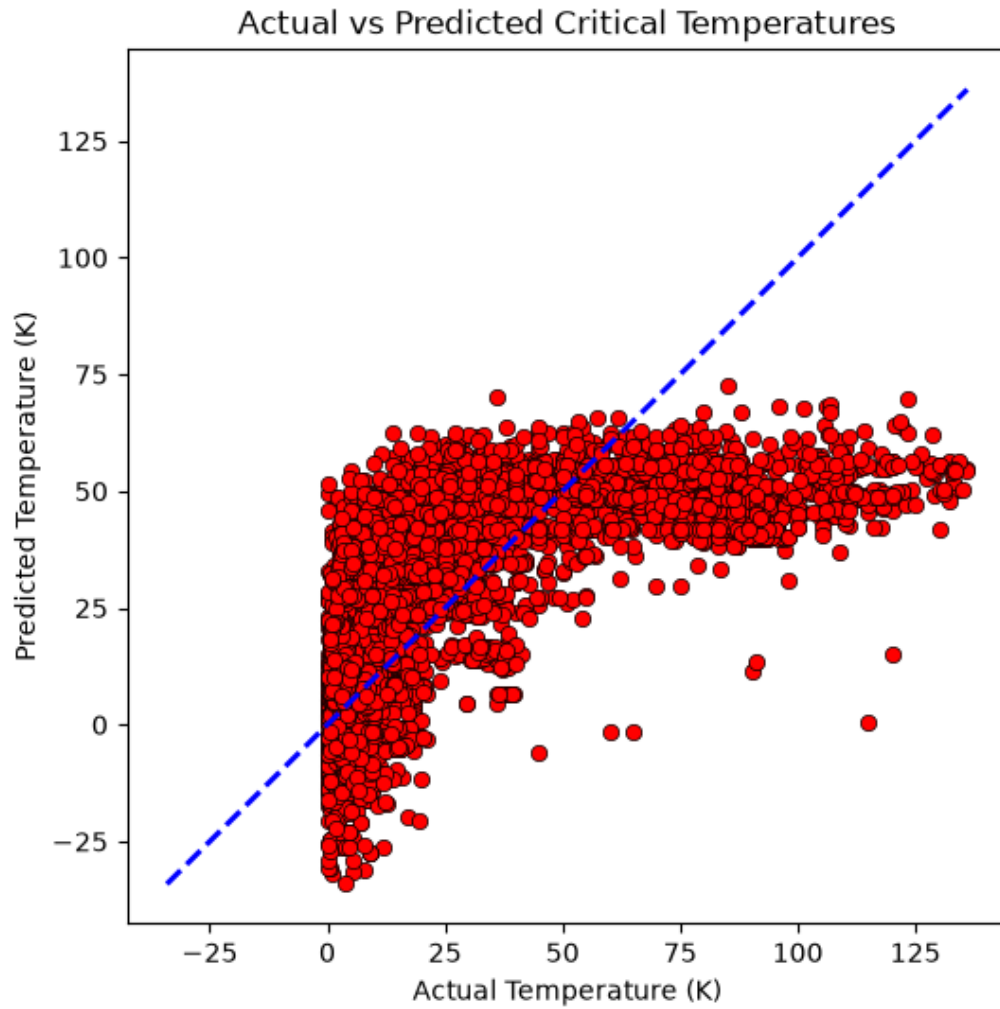


Fig. 3: Actual versus predicted critical temperatures using the selected features.

Although the predictions generally follow the overall trend of the actual values, a considerable amount of scatter can be observed around the ideal prediction line, especially for larger critical temperatures. This indicates that the purely linear model is unable to capture some nonlinear relationships present in the data.

Figure 4 presents the prediction performance after incorporating squared features.

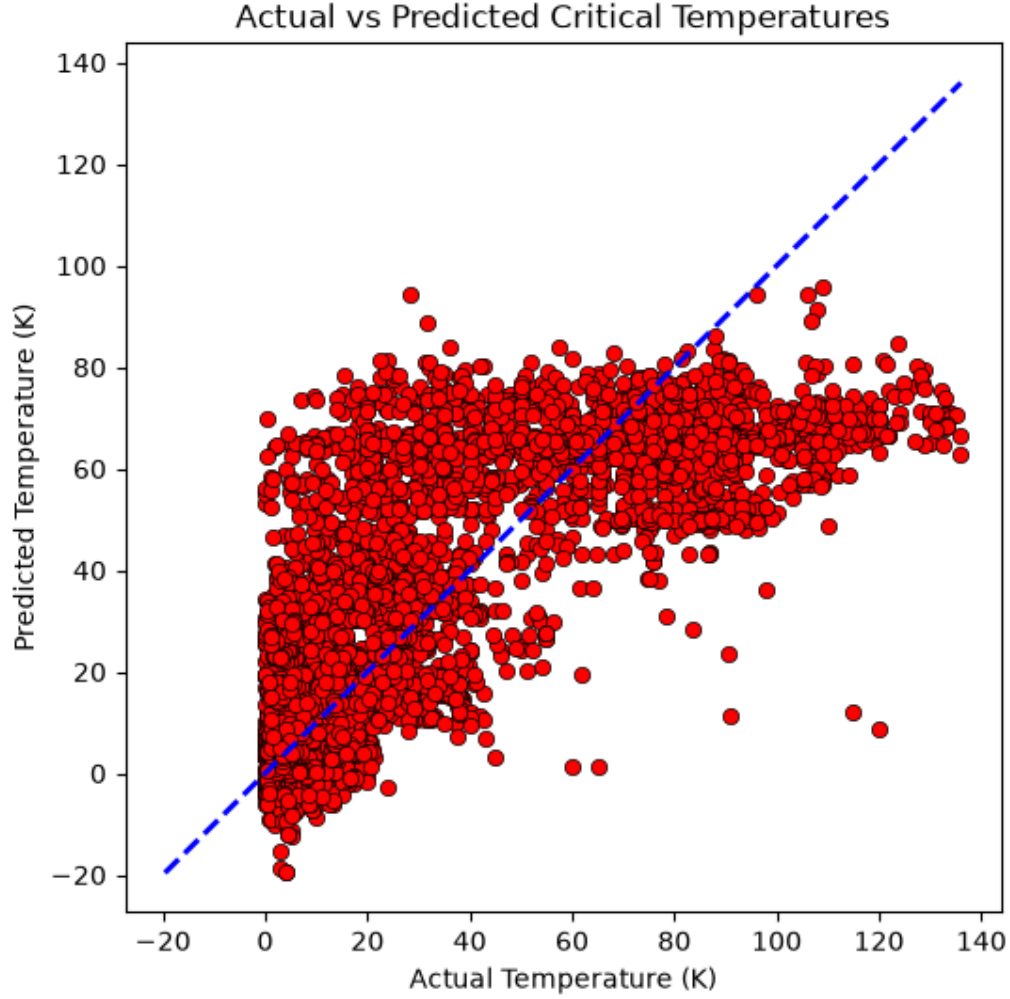


Fig. 4: Actual versus predicted critical temperatures after adding squared features.

Compared with the baseline model, the predicted values are distributed much closer to the ideal diagonal line. The reduction in dispersion indicates that polynomial feature expansion improves the model's ability to approximate the actual critical temperatures.

D. Quantitative Performance Evaluation

To objectively compare the two models, Mean Squared Error (MSE) and the coefficient of determination (R^2) were calculated on the testing dataset.

TABLE I: Performance comparison between the baseline Linear Regression model and the polynomial feature model

Metric	Selected Features	Squared Features	Improvement
Training MSE	646.18	426.06	34.06%
Testing MSE	616.88	423.82	31.30%
R^2 Score	0.4619	0.6303	+0.1684

The results demonstrate that incorporating squared features substantially improves prediction performance.

The training MSE decreases from 646.18 to 426.06, representing an improvement of approximately 34%. Similarly, the testing MSE decreases from 616.88 to 423.82, corresponding to an improvement of approximately 31%.

Furthermore, the coefficient of determination increases from 0.4619 to 0.6303. This indicates that the polynomial feature model explains approximately 63% of the variance in the critical temperature, compared to only 46% for the baseline Linear Regression model.

E. Comparative Analysis

Table I and Figures 3–4 consistently show that feature engineering significantly enhances the predictive capability of the Linear Regression model.

The baseline model captures the overall linear trend but struggles to model complex relationships among the material properties, resulting in higher prediction errors and lower explanatory power.

After adding squared features, the model is able to represent nonlinear interactions between the selected attributes and the superconducting critical temperature. This leads to lower training and testing errors, faster convergence to a smaller cost value, and a noticeable increase in the R^2 score.

Overall, the polynomial feature model achieves superior performance across all evaluation metrics and provides more accurate predictions than the standard Linear Regression model.

F. Evaluation Metrics

Once trained, the model is evaluated against the unseen test dataset. The following regression metrics are used: MSE and RMSE. For this assignment only the MSE is being reported, we calculated the RMSE only for our better understanding.

- **Mean Squared Error (MSE):**

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (1)$$

- **Root Mean Squared Error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

IV. DISCUSSION

The objective of this study was to investigate the effectiveness of Linear Regression for predicting the critical temperature of superconducting materials and to evaluate whether polynomial feature engineering could improve the predictive performance of the model. The experimental results demonstrate that incorporating squared features significantly enhances the model's ability to capture the relationship between material properties and critical temperature.

TABLE II: Performance and Convergence Metrics of the Models

Model Structure	Testing MSE	Testing R^2	Iterations	Final Cost
Baseline Linear Regression	616.88	0.4619	2,000	334
Polynomial Regression (Degree 2)	423.82	0.6303	5,000	214

A. Baseline Model Performance

The baseline Linear Regression model, trained using the selected twenty features, achieved a testing Mean Squared Error (MSE) of 616.88 and an R^2 score of 0.4619. These results indicate that the model is capable of identifying the general trend in the data. The scatter plot of actual versus predicted values further confirms this observation, as considerable dispersion exists around the ideal prediction line, particularly for materials with higher critical temperatures.

B. Impact of Polynomial Feature Engineering

To address this limitation, polynomial feature engineering was applied by introducing squared terms of the selected features. This modification substantially improved the model performance, reducing the testing MSE to 423.82 and increasing the R^2 score to 0.6303. This indicates that the non-linear features allow the model to better represent the complex relationships that exist among the superconducting material properties.

The training process also supports this conclusion. The cost function for the baseline model converged from approximately 1,188.25 to 334 after 2,000 iterations, whereas the polynomial feature model converged from approximately 1,149.99 to nearly 214 after 5,000 iterations. The lower final cost suggests that the polynomial model achieves a better fit to the training data while maintaining improved performance on the testing dataset.

A comparison of the prediction plots reveals that the polynomial model produces predictions that are more closely aligned with the actual critical temperatures. The clustering of points around the diagonal line is noticeably stronger than that of the baseline model, indicating improved prediction accuracy and reduced residual error. This observation is consistent with the improvements observed in both the MSE and R^2 evaluation metrics.

C. Physical Interpretation and Error Analysis

The improvement achieved through polynomial feature expansion can be attributed to the nature of superconductivity data. The critical temperature of a material is influenced by multiple physical and chemical properties that often interact in nonlinear ways. A standard Linear Regression model assumes a purely linear relationship between the input variables and the target variable, which limits its ability to model these complex interactions. By introducing squared features, the model gains additional flexibility and is able to approximate nonlinear patterns without significantly increasing computational complexity.

Although the polynomial model outperforms the baseline Linear Regression model, there is still room for improvement. The achieved R^2 score of 0.6303 indicates that approximately 37% of the variance in the critical temperature remains unexplained. This suggests that higher-order interactions or more sophisticated machine learning algorithms may further improve prediction accuracy.

D. Limitations and Future Work

Future work may explore feature interaction terms, regularized regression methods such as Ridge and Lasso Regression, or non-linear machine learning models including Decision Trees, Random Forests, Support Vector Regression, and Neural Networks. Additionally, advanced feature selection techniques and hyperparameter optimization could further enhance the model's generalization capability.

E. Conclusion

Overall, the experimental findings demonstrate that feature engineering plays a critical role in improving the predictive performance of Linear Regression for superconductivity critical temperature prediction. The addition of squared features resulted in lower training and testing errors, improved convergence behavior, and a significant increase in the coefficient of determination, making the polynomial feature model a more effective approach for this regression task.